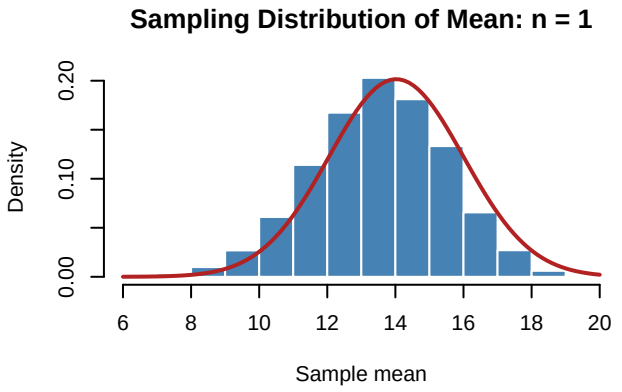
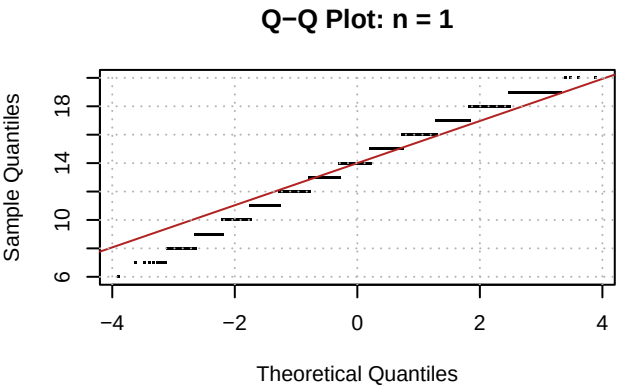
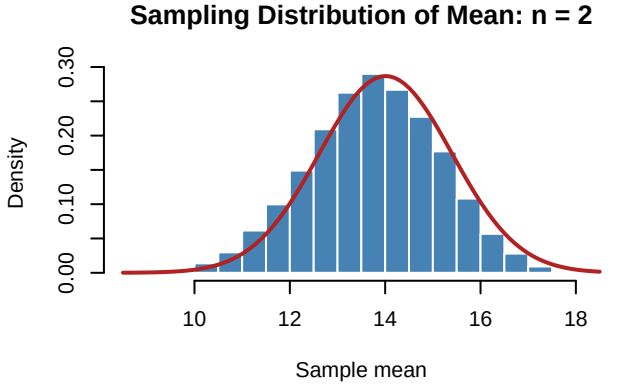
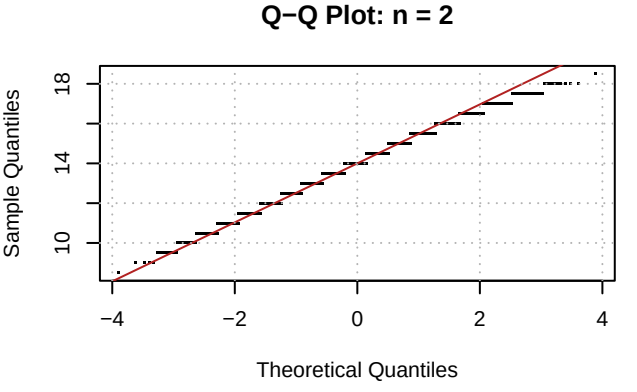
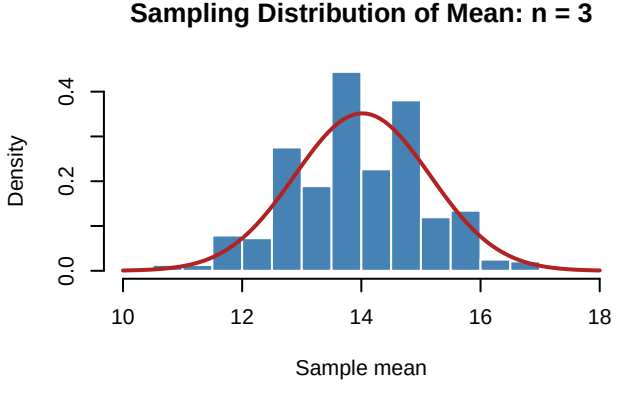
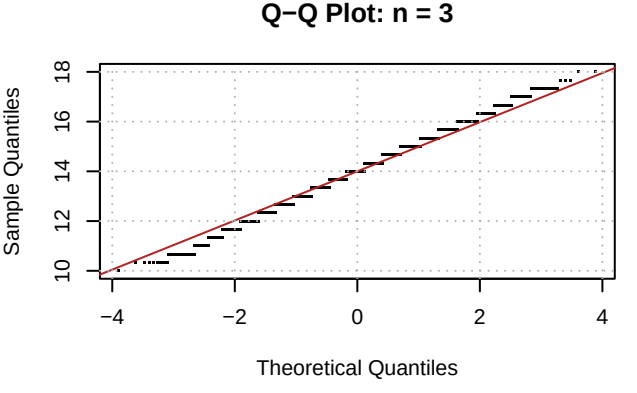

Sampling Distribution of Mean: n = 1
Density
0.00
0.10
0.20
6
8
10
12
14
16
18
20
Sample mean

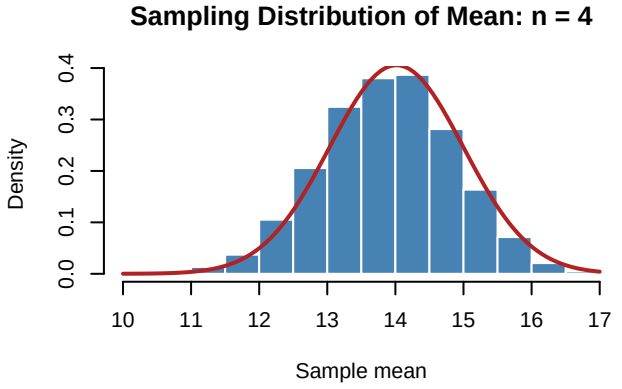

Q−Q Plot: n = 1
Sample Quantiles
6
10
14
18
−4
−2
0
2
4
Theoretical Quantiles

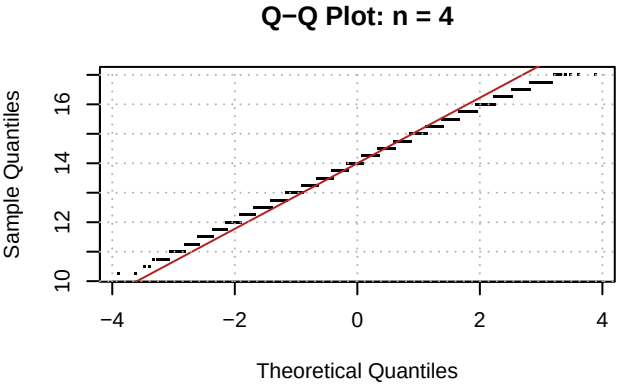

Sampling Distribution of Mean: n = 2
Density
0.00
0.10
0.20
0.30
10
12
14
16
18
Sample mean

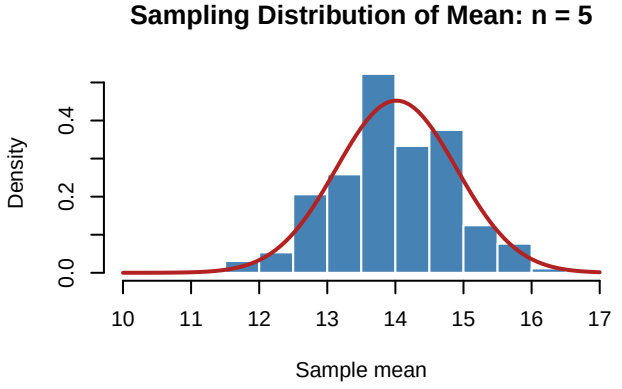

Q−Q Plot: n = 2
Sample Quantiles
10
14
18
−4
−2
0
2
4
Theoretical Quantiles

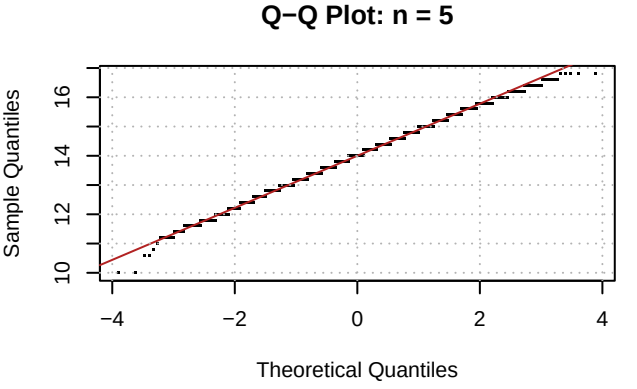

Sampling Distribution of Mean: n = 3
Density
0.0
0.2
0.4
10
12
14
16
18
Sample mean

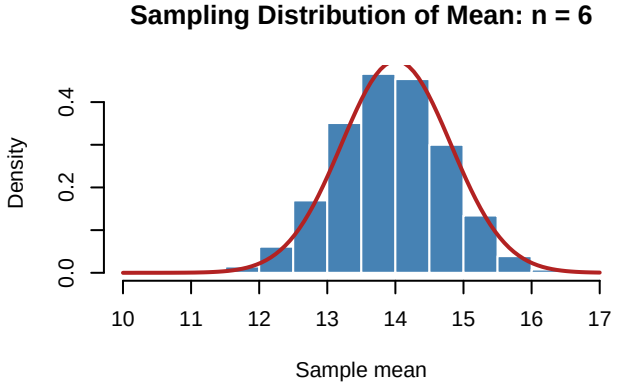

Q−Q Plot: n = 3
Sample Quantiles
10
12
14
16
18
−4
−2
0
2
4
Theoretical Quantiles

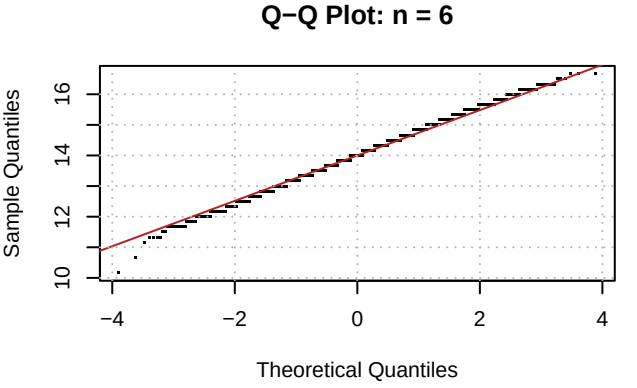

Sampling Distribution of Mean: n = 4
Density
0.0
0.1
0.2
0.3
0.4
10
11
12
13
14
15
16
17
Sample mean

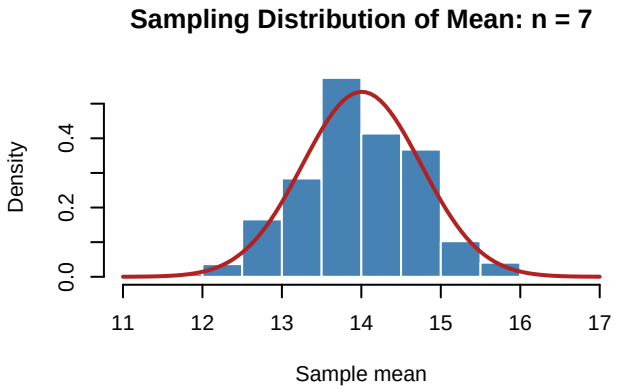

Q−Q Plot: n = 4
Sample Quantiles
10
12
14
16
−4
−2
0
2
4
Theoretical Quantiles

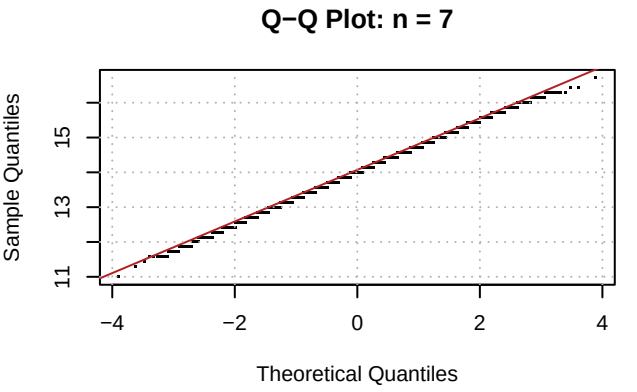

Sampling Distribution of Mean: n = 5
Density
0.0
0.2
0.4
10
11
12
13
14
15
16
17
Sample mean

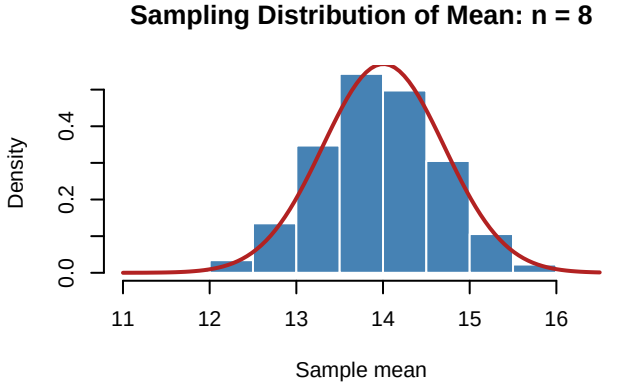

Q−Q Plot: n = 5
Sample Quantiles
10
12
14
16
−4
−2
0
2
4
Theoretical Quantiles

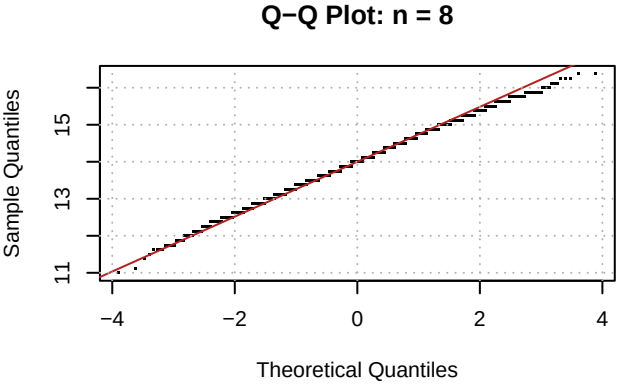

Sampling Distribution of Mean: n = 6
Density
0.0
0.2
0.4
10
11
12
13
14
15
16
17
Sample mean

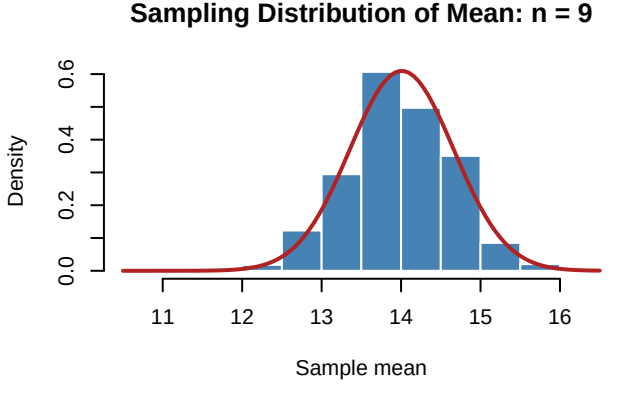

Q−Q Plot: n = 6
Sample Quantiles
10
12
14
16
−4
−2
0
2
4
Theoretical Quantiles

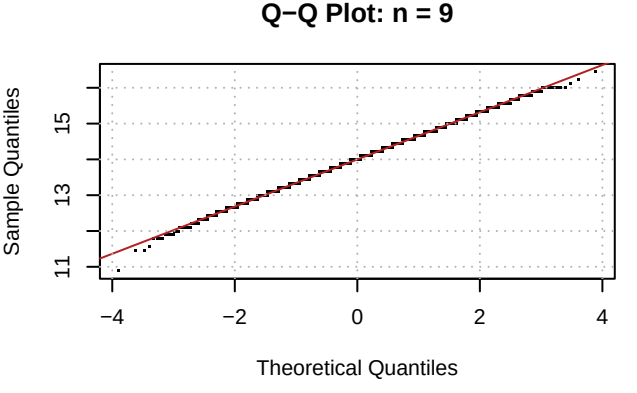

Sampling Distribution of Mean: n = 7
Density
0.0
0.2
0.4
11
12
13
14
15
16
17
Sample mean

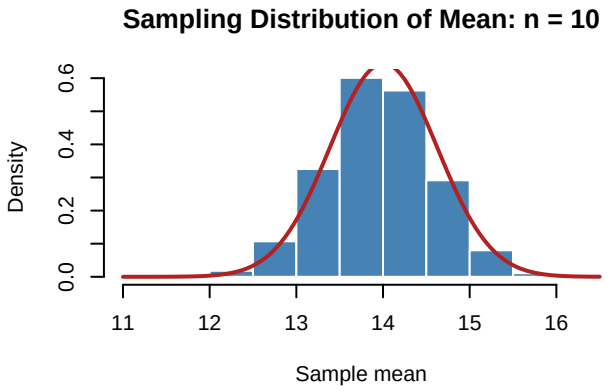

Q−Q Plot: n = 7
Sample Quantiles
11
13
15
−4
−2
0
2
4
Theoretical Quantiles

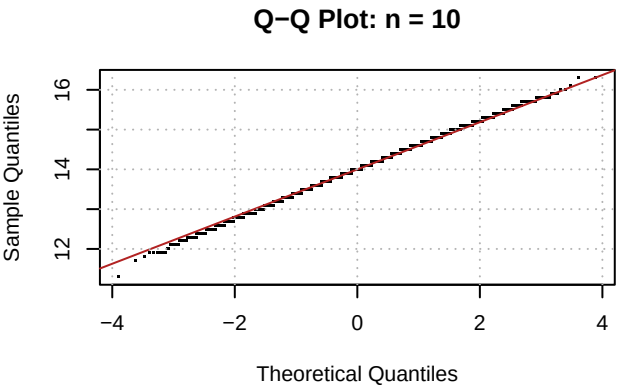

Sampling Distribution of Mean: n = 8
Density
0.0
0.2
0.4
11
12
13
14
15
16
Sample mean

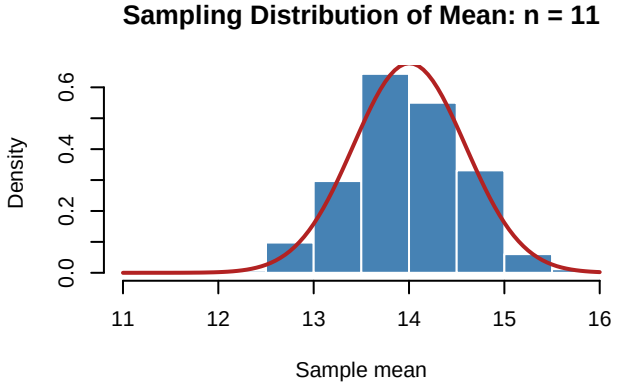

Q−Q Plot: n = 8
Sample Quantiles
11
13
15
−4
−2
0
2
4
Theoretical Quantiles

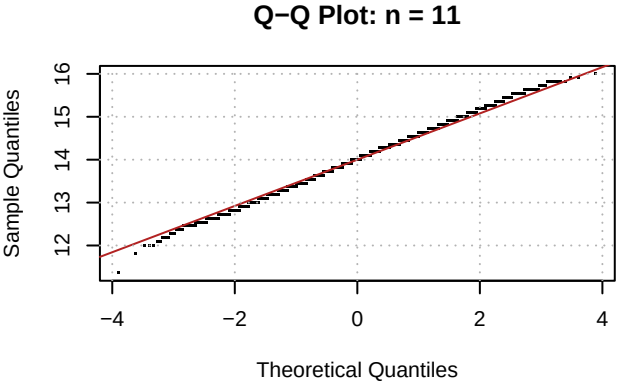

Sampling Distribution of Mean: n = 9
Density
0.0
0.2
0.4
0.6
11
12
13
14
15
16
Sample mean

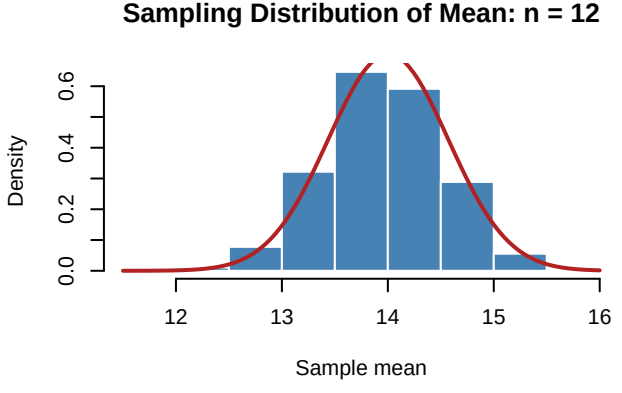

Q−Q Plot: n = 9
Sample Quantiles
11
13
15
−4
−2
0
2
4
Theoretical Quantiles

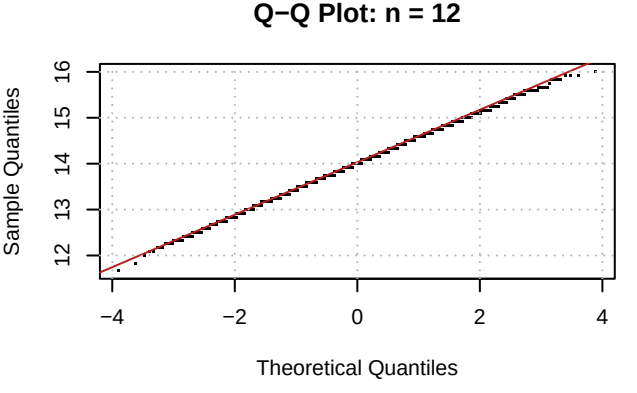

Sampling Distribution of Mean: n = 10
Density
0.0
0.2
0.4
0.6
11
12
13
14
15
16
Sample mean

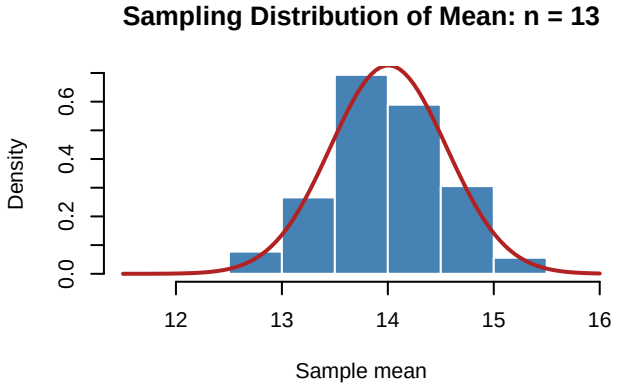

Q−Q Plot: n = 10
Sample Quantiles
12
14
16
−4
−2
0
2
4
Theoretical Quantiles

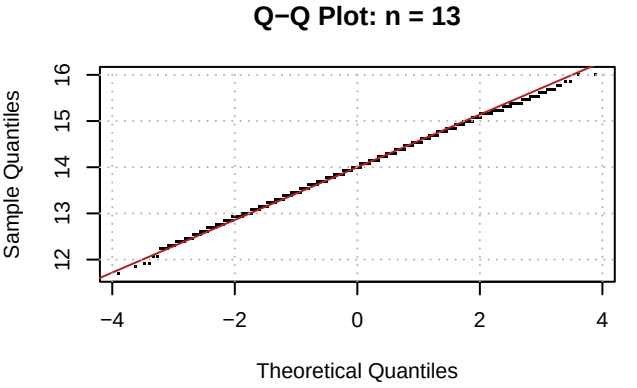

Sampling Distribution of Mean: n = 11
Density
0.0
0.2
0.4
0.6
11
12
13
14
15
16
Sample mean

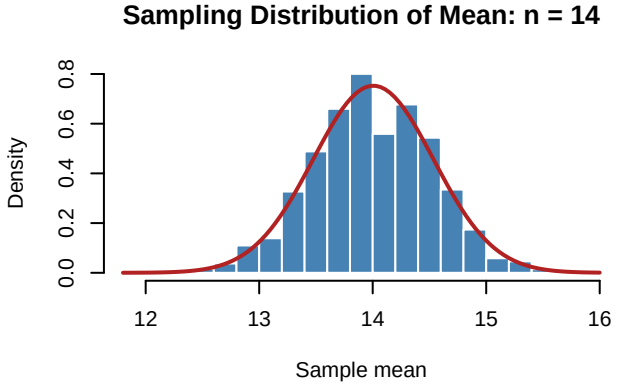

Q−Q Plot: n = 11
Sample Quantiles
12
13
14
15
16
−4
−2
0
2
4
Theoretical Quantiles

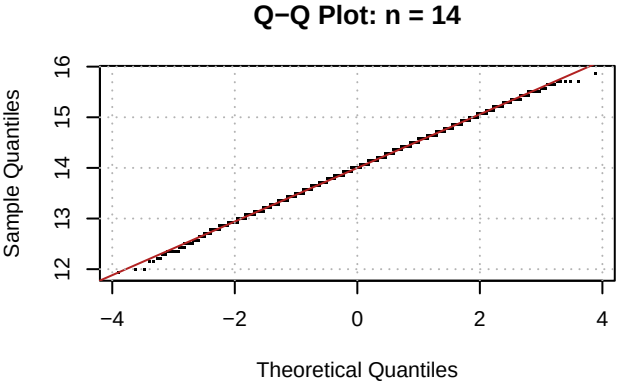

Sampling Distribution of Mean: n = 12
Density
0.0
0.2
0.4
0.6
12
13
14
15
16
Sample mean

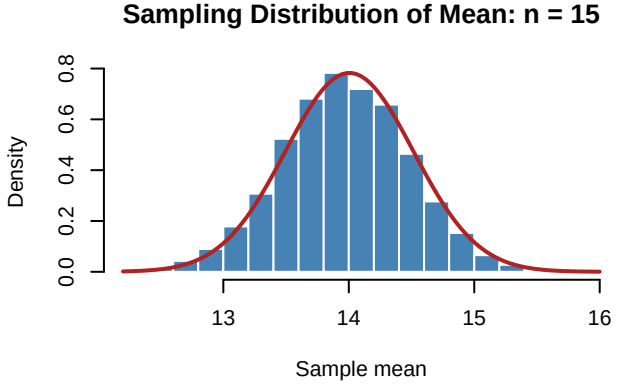

Q−Q Plot: n = 12
Sample Quantiles
12
13
14
15
16
−4
−2
0
2
4
Theoretical Quantiles

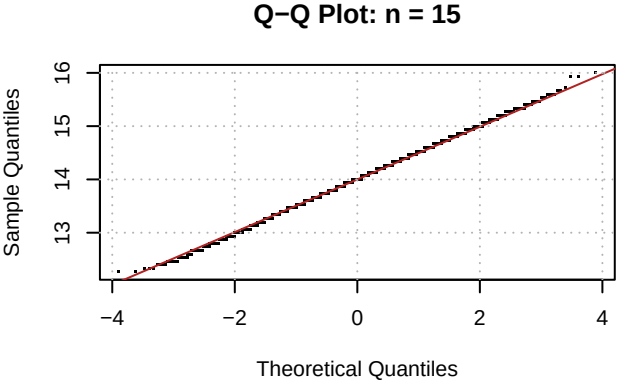

Sampling Distribution of Mean: n = 13
Density
0.0
0.2
0.4
0.6
12
13
14
15
16
Sample mean

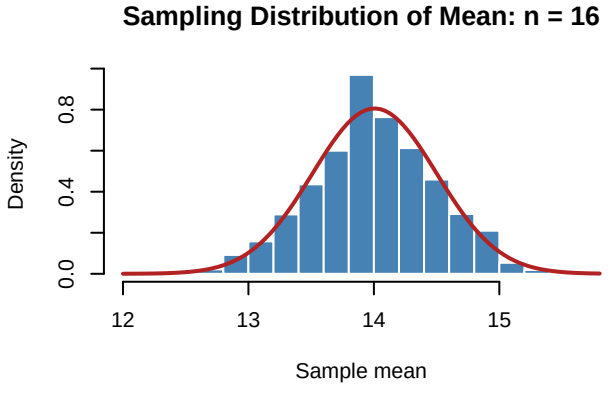

Q−Q Plot: n = 13
Sample Quantiles
12
13
14
15
16
−4
−2
0
2
4
Theoretical Quantiles

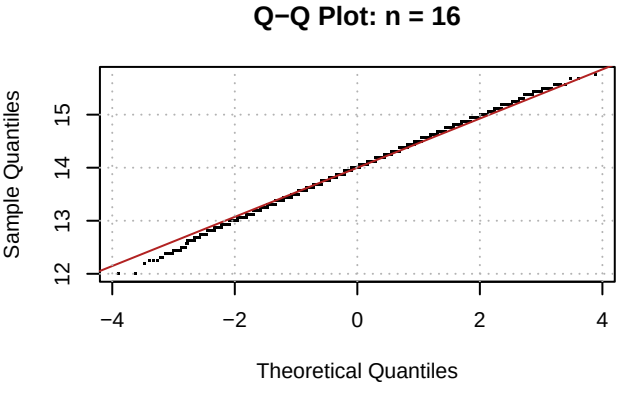

Sampling Distribution of Mean: n = 14
Density
0.0
0.2
0.4
0.6
0.8
12
13
14
15
16
Sample mean

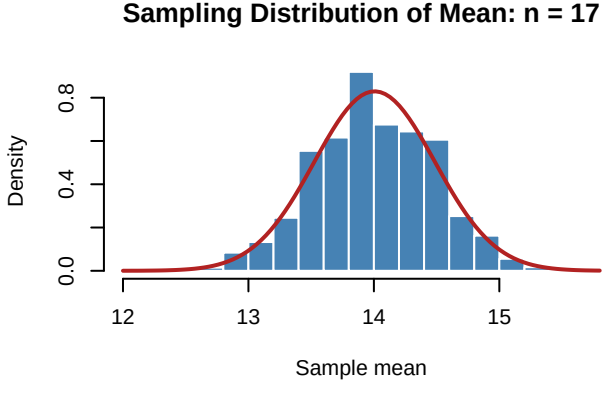

Q−Q Plot: n = 14
Sample Quantiles
12
13
14
15
16
−4
−2
0
2
4
Theoretical Quantiles

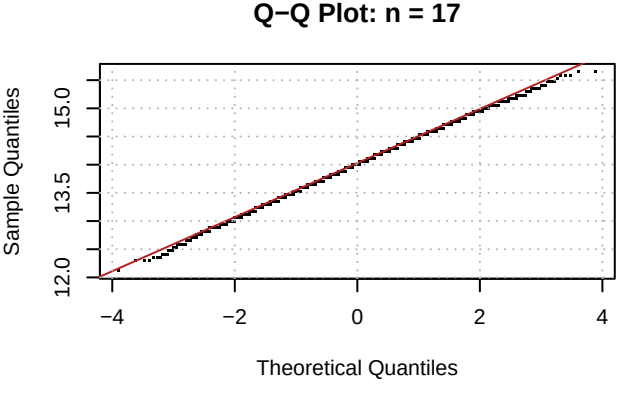

Sampling Distribution of Mean: n = 15
Density
0.0
0.2
0.4
0.6
0.8
13
14
15
16
Sample mean

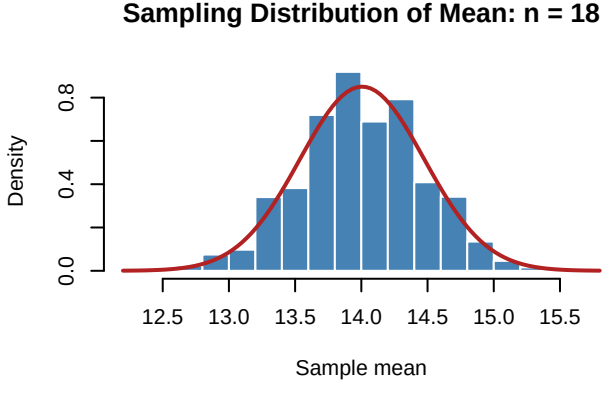

Q−Q Plot: n = 15
Sample Quantiles
13
14
15
16
−4
−2
0
2
4
Theoretical Quantiles

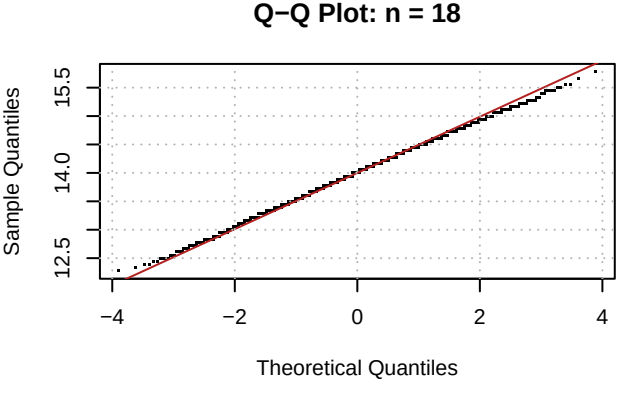

Sampling Distribution of Mean: n = 16
Density
0.0
0.4
0.8
12
13
14
15
Sample mean

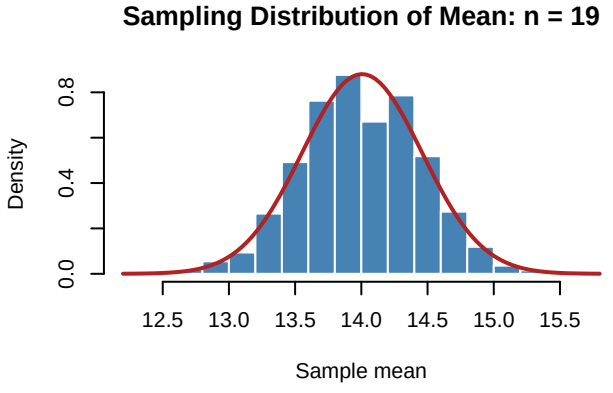

Q−Q Plot: n = 16
Sample Quantiles
12
13
14
15
−4
−2
0
2
4
Theoretical Quantiles

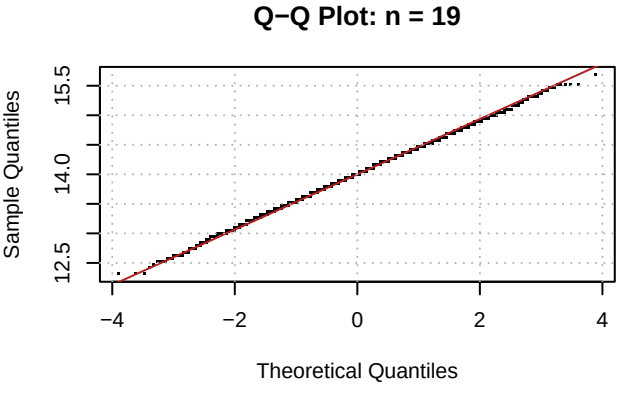

Sampling Distribution of Mean: n = 17
Density
0.0
0.4
0.8
12
13
14
15
Sample mean

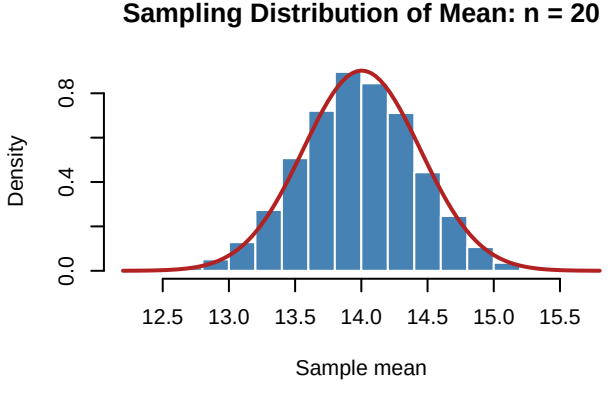

Q−Q Plot: n = 17
Sample Quantiles
12.0
13.5
15.0
−4
−2
0
2
4
Theoretical Quantiles

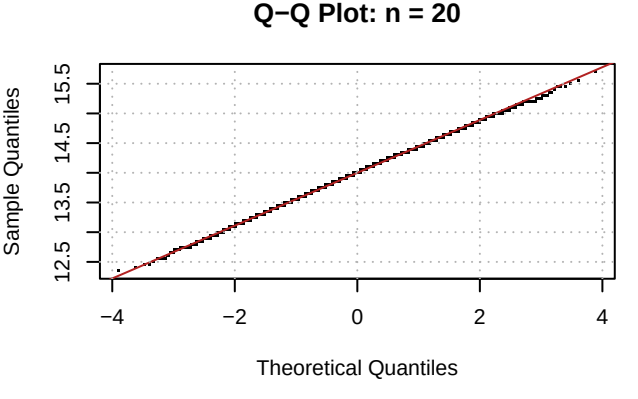

Sampling Distribution of Mean: n = 18
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

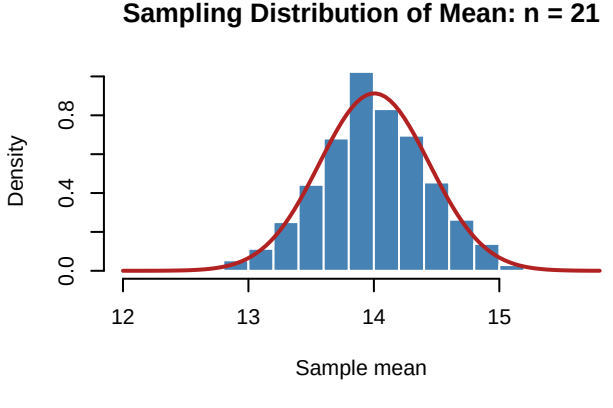

Q−Q Plot: n = 18
Sample Quantiles
12.5
14.0
15.5
−4
−2
0
2
4
Theoretical Quantiles

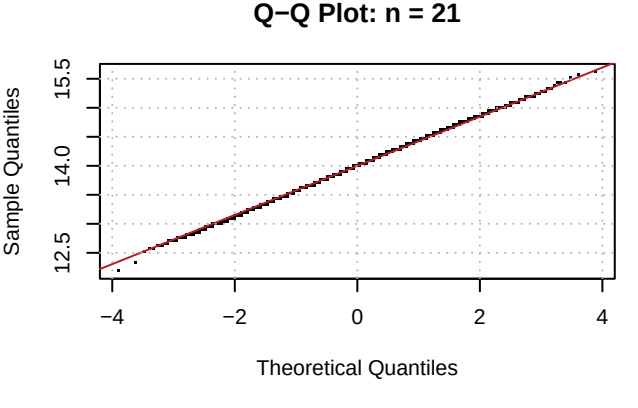

Sampling Distribution of Mean: n = 19
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

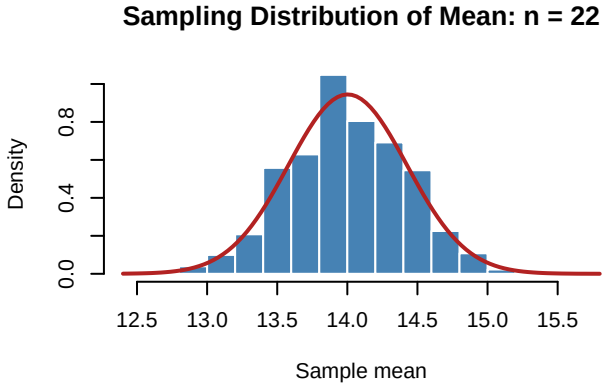

Q−Q Plot: n = 19
Sample Quantiles
12.5
14.0
15.5
−4
−2
0
2
4
Theoretical Quantiles

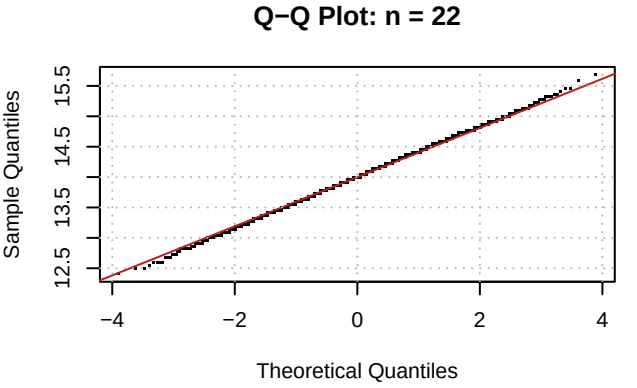

Sampling Distribution of Mean: n = 20
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

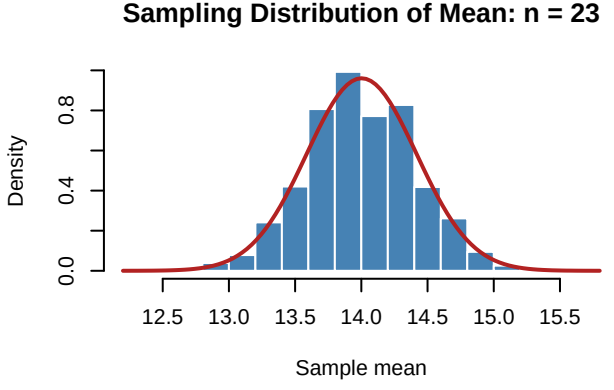

Q−Q Plot: n = 20
Sample Quantiles
12.5
13.5
14.5
15.5
−4
−2
0
2
4
Theoretical Quantiles

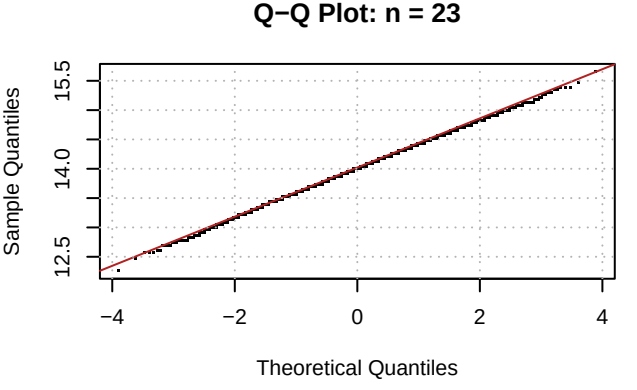

Sampling Distribution of Mean: n = 21
Density
0.0
0.4
0.8
12
13
14
15
Sample mean

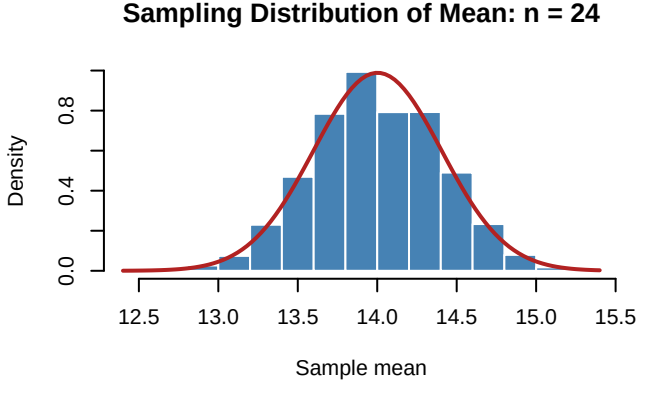

Q−Q Plot: n = 21
Sample Quantiles
12.5
14.0
15.5
−4
−2
0
2
4
Theoretical Quantiles

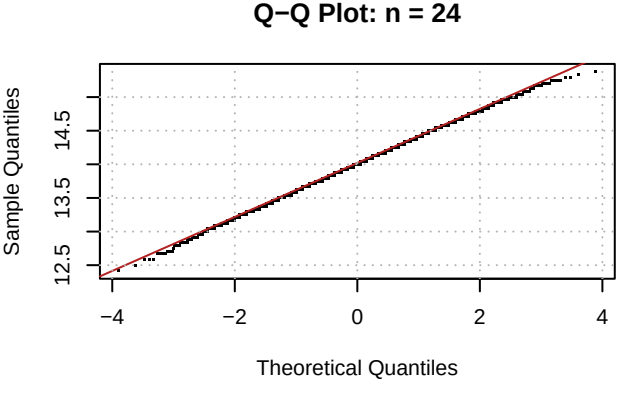

Sampling Distribution of Mean: n = 22
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

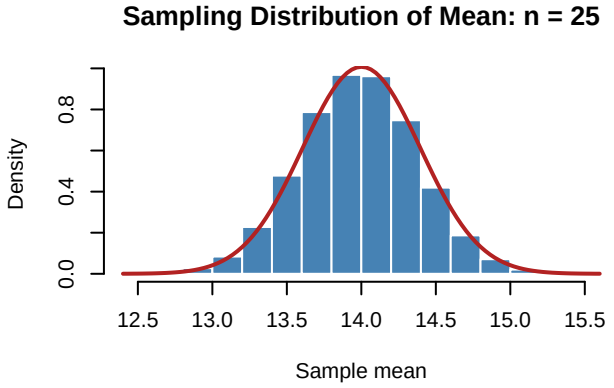

Q−Q Plot: n = 22
Sample Quantiles
12.5
13.5
14.5
15.5
−4
−2
0
2
4
Theoretical Quantiles

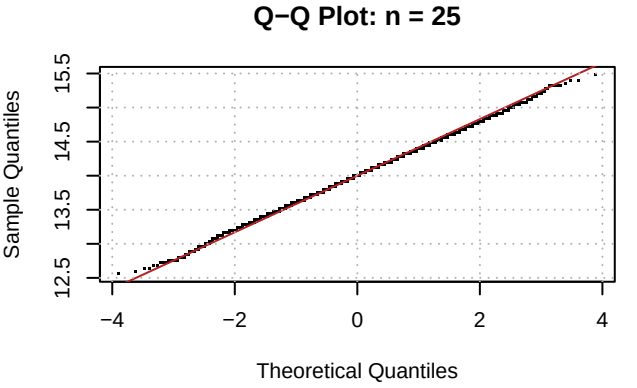

Sampling Distribution of Mean: n = 23
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

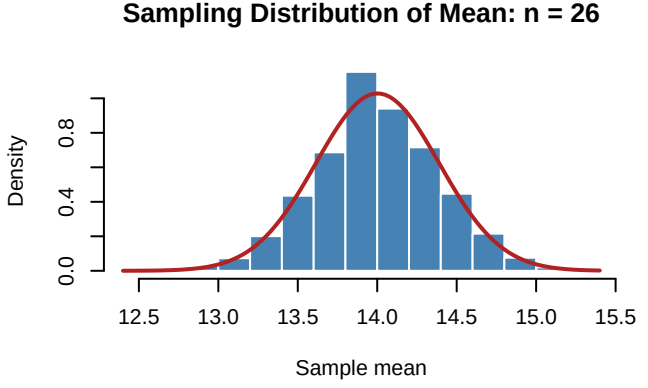

Q−Q Plot: n = 23
Sample Quantiles
12.5
14.0
15.5
−4
−2
0
2
4
Theoretical Quantiles

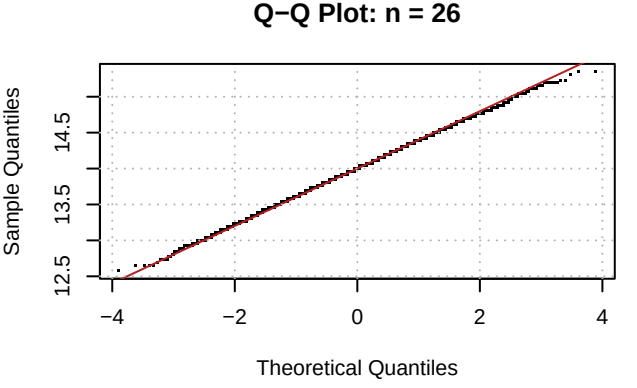

Sampling Distribution of Mean: n = 24
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

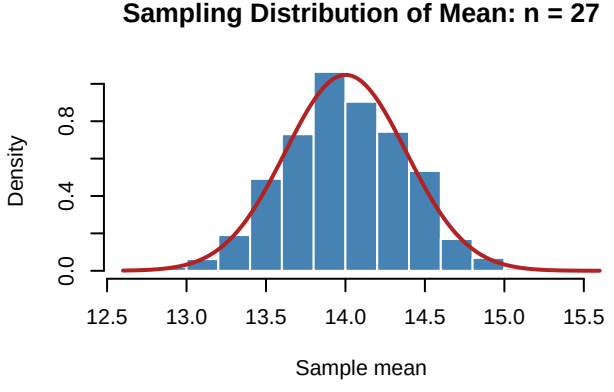

Q−Q Plot: n = 24
Sample Quantiles
12.5
13.5
14.5
−4
−2
0
2
4
Theoretical Quantiles

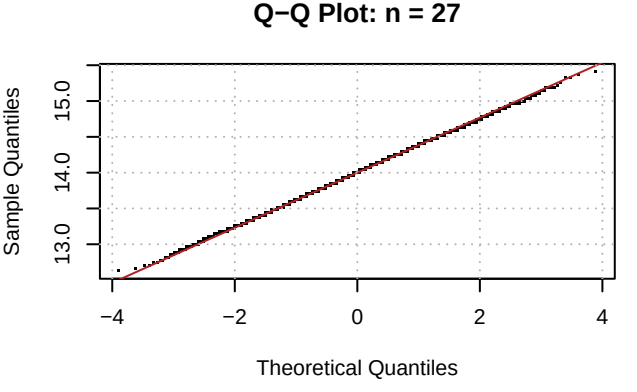

Sampling Distribution of Mean: n = 25
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

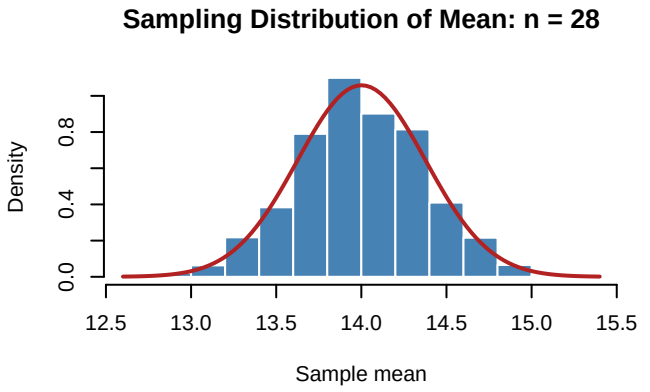

Q−Q Plot: n = 25
Sample Quantiles
12.5
13.5
14.5
15.5
−4
−2
0
2
4
Theoretical Quantiles

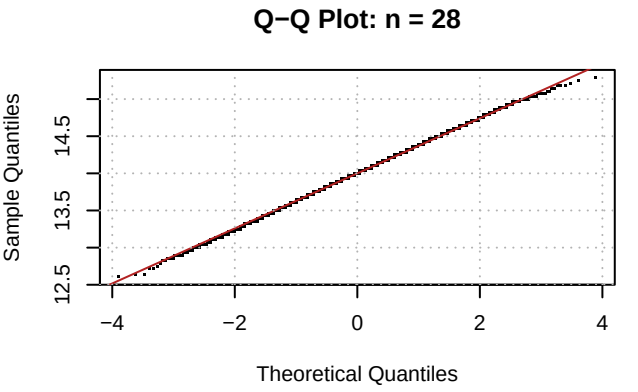

Sampling Distribution of Mean: n = 26
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

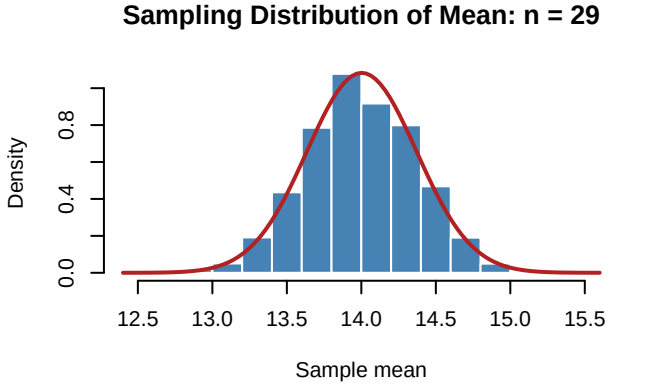

Q−Q Plot: n = 26
Sample Quantiles
12.5
13.5
14.5
−4
−2
0
2
4
Theoretical Quantiles

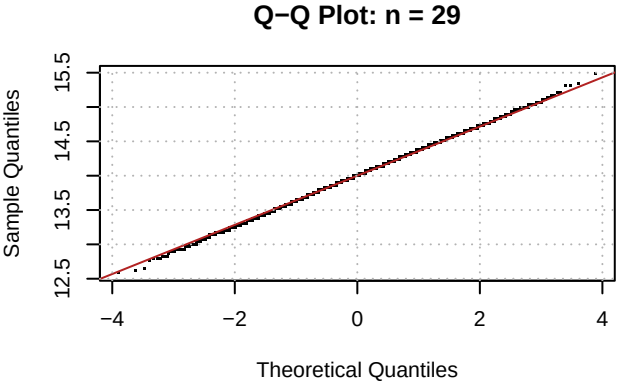

Sampling Distribution of Mean: n = 27
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

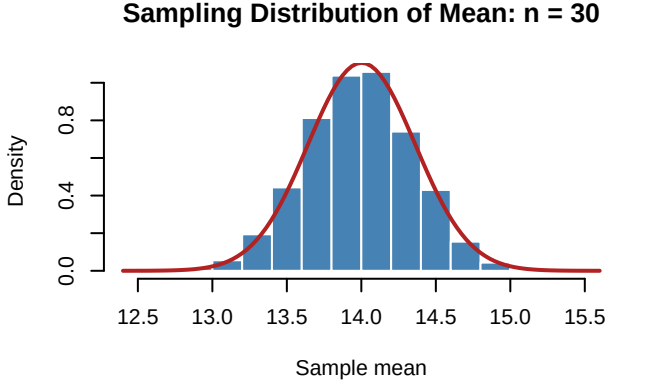

Q−Q Plot: n = 27
Sample Quantiles
13.0
14.0
15.0
−4
−2
0
2
4
Theoretical Quantiles

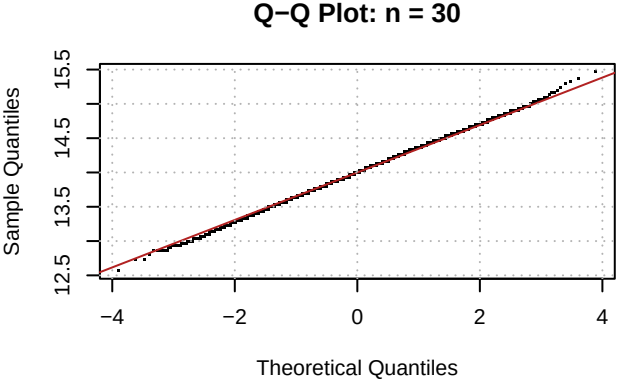

Sampling Distribution of Mean: n = 28
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

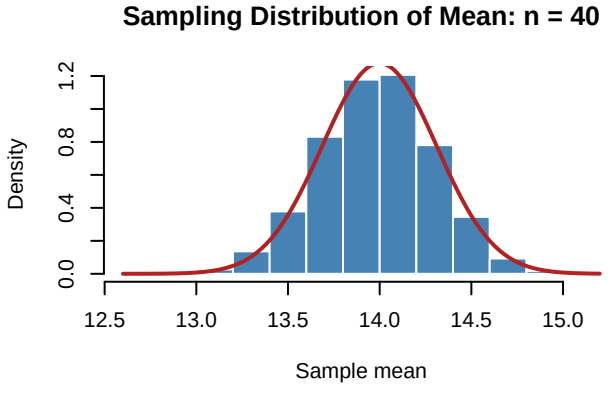

Q−Q Plot: n = 28
Sample Quantiles
12.5
13.5
14.5
−4
−2
0
2
4
Theoretical Quantiles

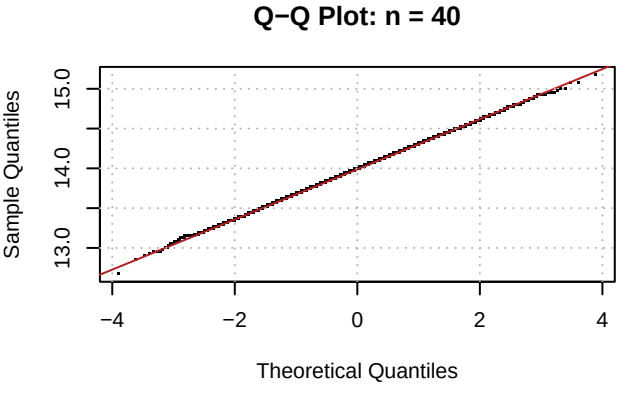

Sampling Distribution of Mean: n = 29
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

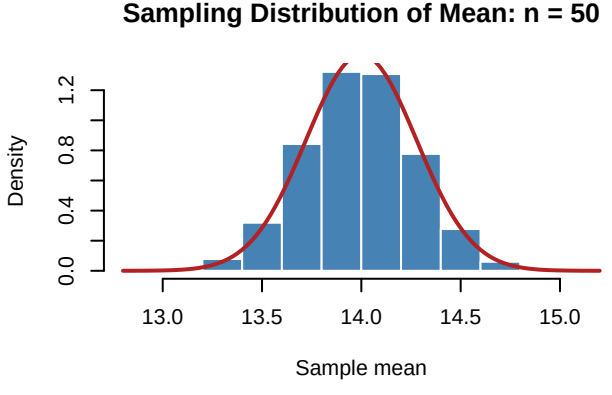

Q−Q Plot: n = 29
Sample Quantiles
12.5
13.5
14.5
15.5
−4
−2
0
2
4
Theoretical Quantiles

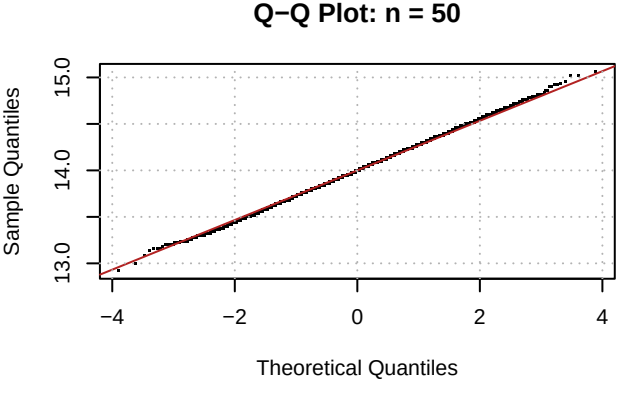

Sampling Distribution of Mean: n = 30
Density
0.0
0.4
0.8
12.5
13.0
13.5
14.0
14.5
15.0
15.5
Sample mean

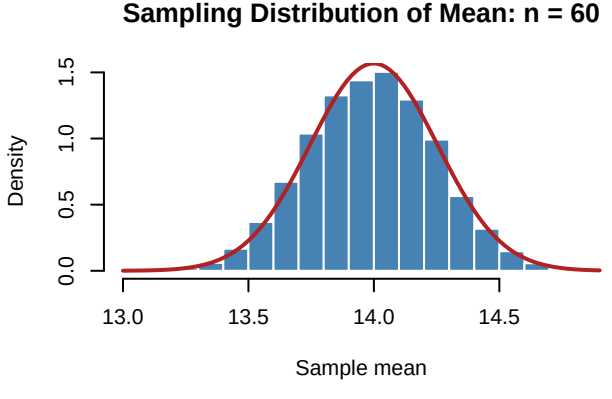

Q−Q Plot: n = 30
Sample Quantiles
12.5
13.5
14.5
15.5
−4
−2
0
2
4
Theoretical Quantiles

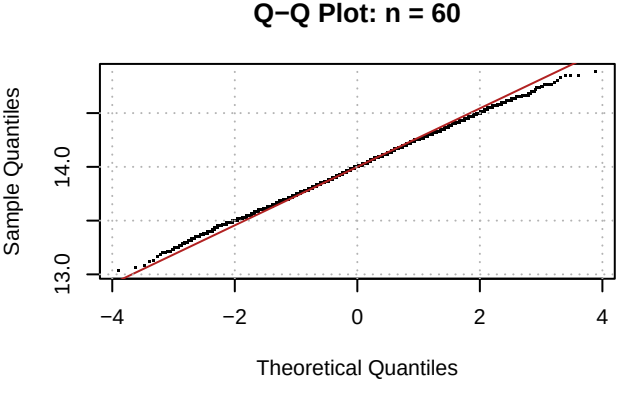

Sampling Distribution of Mean: n = 40
Density
0.0
0.4
0.8
1.2
12.5
13.0
13.5
14.0
14.5
15.0
Sample mean

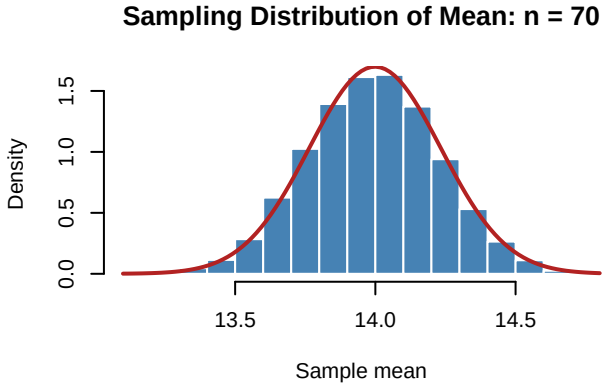

Q−Q Plot: n = 40
Sample Quantiles
13.0
14.0
15.0
−4
−2
0
2
4
Theoretical Quantiles

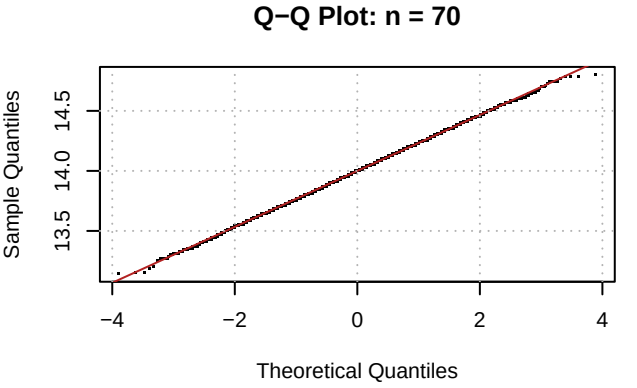

Sampling Distribution of Mean: n = 50
Density
0.0
0.4
0.8
1.2
13.0
13.5
14.0
14.5
15.0
Sample mean

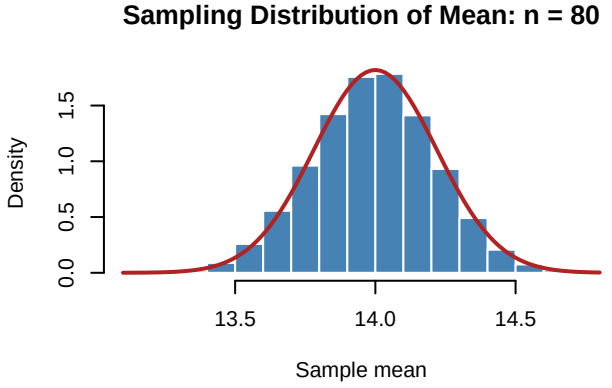

Q−Q Plot: n = 50
Sample Quantiles
13.0
14.0
15.0
−4
−2
0
2
4
Theoretical Quantiles

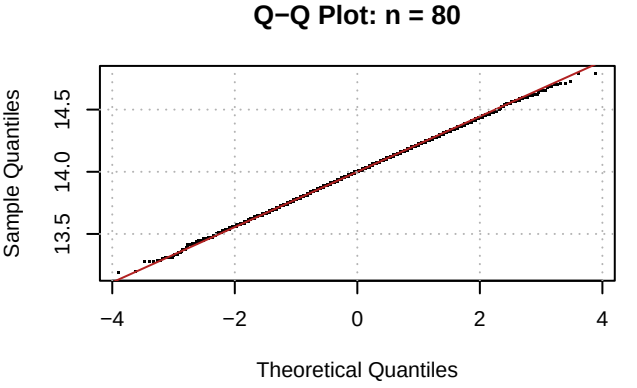

Sampling Distribution of Mean: n = 60
Density
0.0
0.5
1.0
1.5
13.0
13.5
14.0
14.5
Sample mean

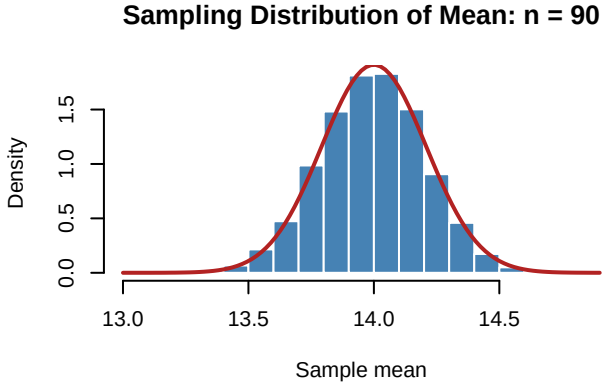

Q−Q Plot: n = 60
Sample Quantiles
13.0
14.0
−4
−2
0
2
4
Theoretical Quantiles

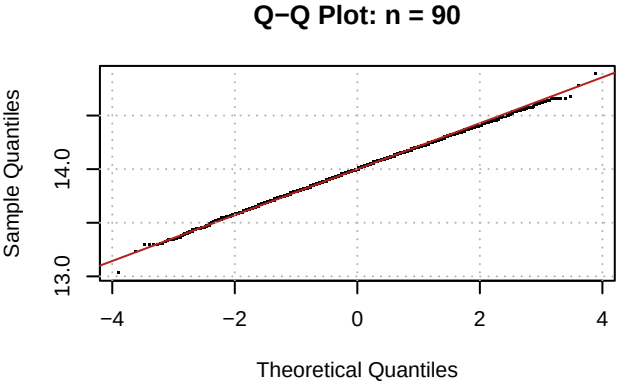

Sampling Distribution of Mean: n = 70
Density
0.0
0.5
1.0
1.5
13.5
14.0
14.5
Sample mean

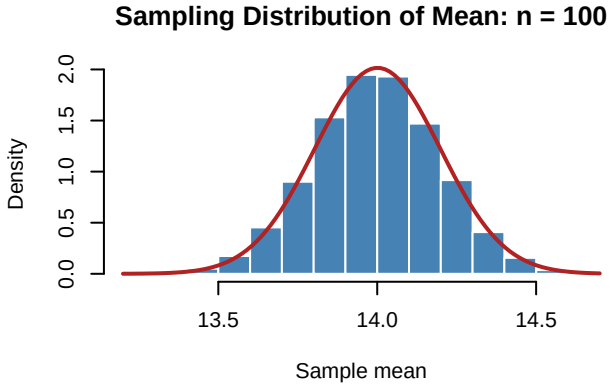

Q−Q Plot: n = 70
Sample Quantiles
13.5
14.0
14.5
−4
−2
0
2
4
Theoretical Quantiles

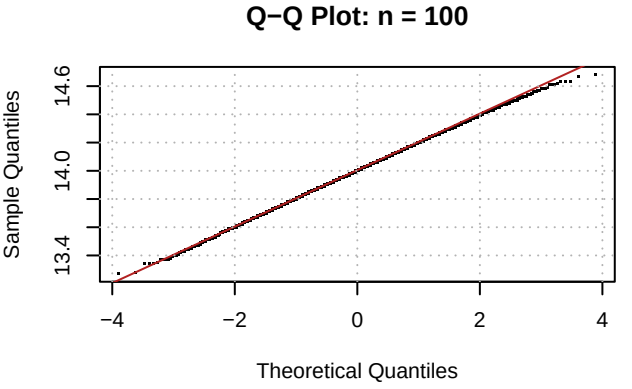

Sampling Distribution of Mean: n = 80
Density
0.0
0.5
1.0
1.5
13.5
14.0
14.5
Sample mean

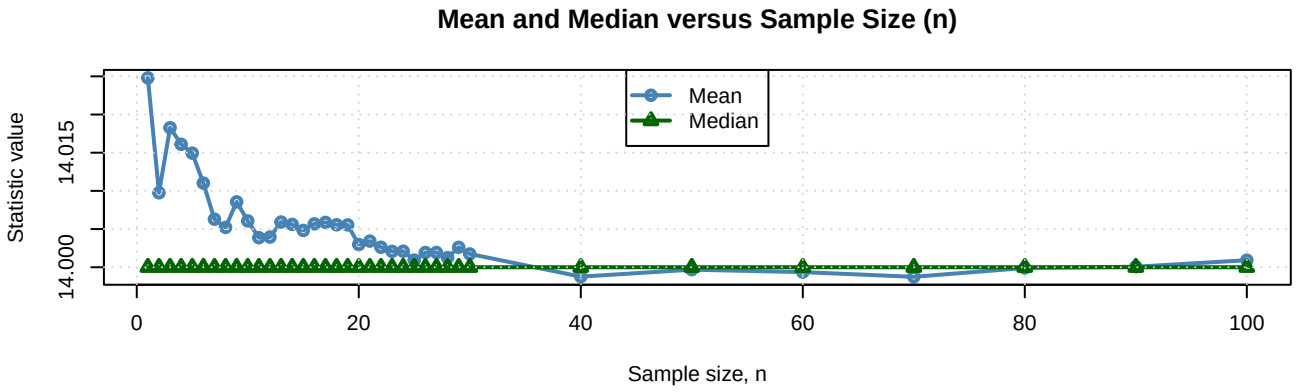

Q−Q Plot: n = 80
Sample Quantiles
13.5
14.0
14.5
−4
−2
0
2
4
Theoretical Quantiles

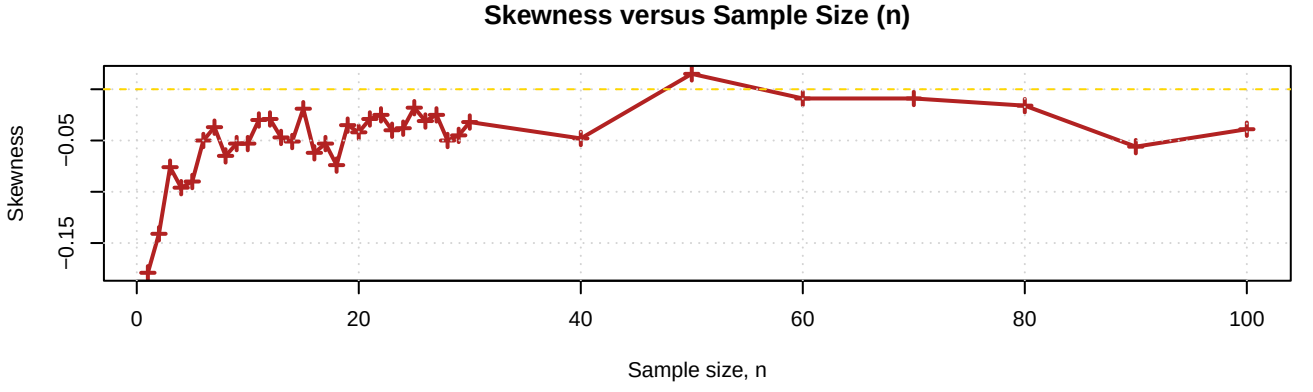

Sampling Distribution of Mean: n = 90
Density
0.0
0.5
1.0
1.5
13.0
13.5
14.0
14.5
Sample mean

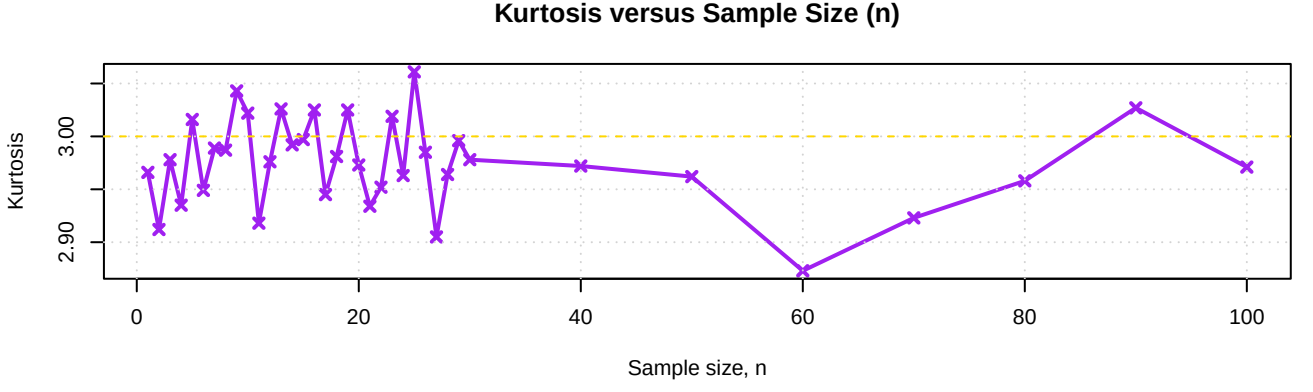

Q−Q Plot: n = 90
Sample Quantiles
13.0
14.0
−4
−2
0
2
4
Theoretical Quantiles


Sampling Distribution of Mean: n = 100
Density
0.0
0.5
1.0
1.5
2.0
13.5
14.0
14.5
Sample mean


Q−Q Plot: n = 100
Sample Quantiles
13.4
14.0
14.6
−4
−2
0
2
4
Theoretical Quantiles


Mean and Median versus Sample Size (n)
Mean
Median
Statistic value
14.000
14.015
0
20
40
60
80
100
Sample size, n


Skewness versus Sample Size (n)
Skewness
−0.15
−0.05
0
20
40
60
80
100
Sample size, n


Kurtosis versus Sample Size (n)
Kurtosis
2.90
3.00
0
20
40
60
80
100
Sample size, n